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Tutorial Course for Intro to Sensor-Integrated Systems

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2026

Core Principles and Physical Modeling of Inductive Sensors

Core Idea: Convert the measured physical quantity (such as displacement $\Delta\delta$) into a change in the reluctance (R_m) of the magnetic circuit, which in turn causes a change in the coil inductance (L).

1. Reluctance Formula:

$$R_m = \frac{l}{\mu S}$$

(Note: The magnetic permeability of the air gap is extremely low, so the total reluctance is mainly determined by the air gap)

2. Inductance Definition Formula:

$$L = \frac{W^2}{R_m}$$

3. Initial Inductance Formula for Variable Air-Gap Inductive Sensors:

$$L_0 = \frac{\mu_0 S_0 W^2}{2\delta_0}$$

Symbols List

- L_0 : Initial inductance, unit: H (Henry) or mH .
- W : Number of coil turns, dimensionless.
- μ_0 : Vacuum permeability, constant $4\pi \times 10^{-7} H/m$.
- S_0 : Cross-sectional area of magnetic circuit/air gap, unit must be converted to: m^2 (if given in cm^2 in the problem, multiply by 10^{-4}).
- δ_0 : Initial air gap length, unit must be converted to: m (if given in cm or mm in the problem, multiply by 10^{-2} or 10^{-3}).

Single-ended vs. Differential Structure

Sensitivity

- **Definition:** The amount of inductance change caused by a unit change in displacement.
- **Formula:**

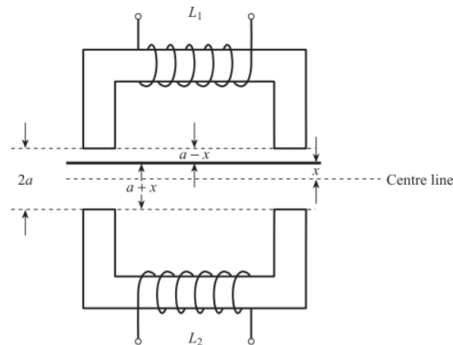
$$K = \frac{\Delta L}{\Delta \delta}$$

Single-ended Structure

- **Characteristics:** The armature movement changes the air gap on one side, resulting in non-linear errors in the output characteristics.
- **Approximate Sensitivity:** Under micro-displacement, it is approximated as $K_{single} \approx \frac{L_0}{\delta_0}$.

Differential Structure

- **Principle:** Two identical coils are used. When the armature moves, one air gap increases ($\delta_0 + \Delta\delta$) while the other air gap decreases by an equal amount ($\delta_0 - \Delta\delta$).
- **Advantage 1 (Improved Linearity):** The inductance change of the two coils subtracts from each other, canceling out the even-harmonic non-linear terms and significantly improving linearity.
- **Advantage 2 (Doubled Sensitivity):** The inductance change of the differential output is $\Delta L_{diff} = 2L_0 \frac{\Delta\delta}{\delta_0}$. This means the differential sensitivity is twice that of the single-ended structure ($K_{diff} = 2K_{single}$).



Exercise

For a variable air-gap inductive sensor, the cross-sectional area of the iron core is $S = 1.5 \text{ cm}^2$, the length of the magnetic circuit is $L = 20 \text{ cm}$, the relative permeability is $\mu_r = 5000$, the initial air gap is $\delta_0 = 0.5 \text{ cm}$, the change in the air gap is $\Delta\delta = \pm 0.1 \text{ mm}$, the permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$, and the number of coil turns is $W = 3000$.

Calculate the sensitivity $\Delta L / \Delta\delta$ of the single-ended sensor. If it is constructed as a differential structure, how will its sensitivity change?

Solution

Step 1: Calculate the initial inductance (L_0)

First, convert the units to the standard SI system ($1 \text{ cm}^2 = 10^{-4} \text{ m}^2$, $1 \text{ cm} = 10^{-2} \text{ m}$):

$$L_0 = \frac{\mu_0 S W^2}{2 \delta_0}$$

$$L_0 = \frac{4\pi \times 10^{-7} \times 1.5 \times 10^{-4} \times 3000^2}{2 \times 0.5 \times 10^{-2}}$$

$$L_0 \approx 169.6 \text{ mH}$$

Step 2: Calculate the inductance after the air gap changes (L)

Note that $\Delta \delta = \pm 0.1 \text{ mm} = \pm 0.01 \text{ cm}$.

$$L = \frac{\mu_0 S W^2}{2 \delta}$$

$$L = \frac{4\pi \times 10^{-7} \times 1.5 \times 10^{-4} \times 3000^2}{2 \times (0.5 \pm 0.01) \times 10^{-2}}$$

$$L = 169.6 \pm 3.5 \text{ mH}$$

Therefore, the change in inductance $\Delta L = 3.5 \text{ mH}$.

Step 3: Calculate the sensitivity of the single-ended sensor (K)

$$K = \frac{\Delta L}{\Delta \delta} = \frac{3.5 \times 10^{-3}}{0.1 \times 10^{-3}} = 35 \text{ H/m}$$

Step 4: Calculate the sensitivity for a differential structure

If connected in a differential structure, the sensitivity is doubled:

$$K_{diff} = 2 \cdot \frac{\Delta L}{\Delta \delta} = 2 \times 35 = 70 \text{ H/m}$$

Exercise

A variable reluctance sensor consists of a core, a variable air gap, and an armature. The core is a steel rod of diameter 1 cm bent into a semi-circle of mean diameter 4 cm. A coil of 500 turns is wound onto the core. The armature is a steel plate of thickness 0.5 cm. The relative permeability of the steel is 100. Calculate the inductance of the sensor for air gaps of 1 mm and 3 mm.

Solution

Step 1: Identify and calculate the physical parameters of the magnetic circuit

Core (Steel rod):

- Area $A_c = \frac{\pi d^2}{4} = \frac{\pi \times (0.01 \text{ m})^2}{4} = 7.854 \times 10^{-5} \text{ m}^2$
- Path length $l_c = \frac{\pi D}{2} = \frac{\pi \times (0.04 \text{ m})}{2} = 0.06283 \text{ m}$

Armature (Steel plate):

- Assuming the effective width matches the core diameter (1 cm) to cover the poles, the cross-sectional area is $A_a = \text{width} \times \text{thickness} = 0.01 \text{ m} \times 0.005 \text{ m} = 5 \times 10^{-5} \text{ m}^2$
- Path length between poles $l_a = 0.04 \text{ m}$

Air Gaps:

- There are two air gaps in series. For an air gap of x , the total gap length is $l_g = 2x$.
- The effective area is the pole face area: $A_g = A_c = 7.854 \times 10^{-5} \text{ m}^2$

Step 2: Calculate the magnetic reluctance (R_m) of the iron parts

The general formula for magnetic reluctance is

$$R_m = \frac{l}{\mu_0 \mu_r A}$$

(where $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$).

- Core reluctance:** $R_{mc} = \frac{0.06283}{4\pi \times 10^{-7} \times 100 \times 7.854 \times 10^{-5}} = 6.366 \times 10^6 \text{ H}^{-1}$

- Armature reluctance:**

$$R_{ma} = \frac{0.04}{4\pi \times 10^{-7} \times 100 \times 5 \times 10^{-5}} = 6.366 \times 10^6 \text{ H}^{-1}$$

- Total iron reluctance:**

$$R_{iron} = R_{mc} + R_{ma} = 12.732 \times 10^6 \text{ H}^{-1}$$

Step 3: Calculate the inductance for an air gap of $x = 1 \text{ mm}$

- Air gap reluctance** (where $x = 0.001 \text{ m}$ and $\mu_r = 1$ for air):

$$R_{mg} = \frac{2 \times 0.001}{4\pi \times 10^{-7} \times 1 \times 7.854 \times 10^{-5}} = 20.264 \times 10^6 \text{ H}^{-1}$$

- Total reluctance:**

$$R_{total} = R_{iron} + R_{mg} = (12.732 + 20.264) \times 10^6 = 32.996 \times 10^6 \text{ H}^{-1}$$

- Inductance ($L = \frac{N^2}{R_{total}}$):**

$$L = \frac{500^2}{32.996 \times 10^6} \approx 0.00758 \text{ H} = \mathbf{7.58 \text{ mH}}$$

Solution

Step 4: Calculate the inductance for an air gap of $x = 3$ mm

- Air gap reluctance (where $x = 0.003$ m):

$$R_{mg} = \frac{2 \times 0.003}{4\pi \times 10^{-7} \times 1 \times 7.854 \times 10^{-5}} = 60.792 \times 10^6 \text{ H}^{-1}$$

- Total reluctance:

$$R_{total} = R_{iron} + R_{mg} = (12.732 + 60.792) \times 10^6 = 73.524 \times 10^6 \text{ H}^{-1}$$

- Inductance ($L = \frac{N^2}{R_{total}}$):

$$L = \frac{500^2}{73.524 \times 10^6} \approx 0.00340 \text{ H} = \mathbf{3.40 \text{ mH}}$$

Core Essentials of Capacitive Sensors

Basic Physical Modeling and Three Major Classifications:

➤ Core Formula:

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

➤ Three Major Variant Types (Determined by the three elements in the formula):

- **Variable Distance Type (Variable d):** The output is inversely proportional to the plate distance (non-linear). In engineering, a differential structure must be adopted to improve linearity.
- **Variable Area Type (Variable A):** The output is strictly linear with the area, commonly used to measure larger linear or angular displacements.
- **Variable Dielectric Constant Type (Variable ϵ_r):** The output is strictly linear with the change in dielectric medium, a typical application is the capacitive liquid level gauge.

Classic Application Modeling: Cylindrical Capacitive Liquid Level Gauge

- **Modeling Idea:** The liquid height is h , which can be equivalent to a parallel combination of the liquid-segment capacitance and the air-segment capacitance.
- **Derivation Conclusion:**

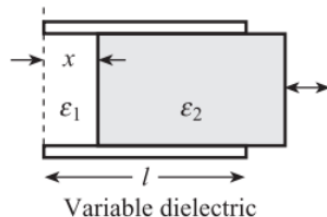
$$C_{total} = C_{liquid} + C_{air} = C_0 + K \cdot h$$

- This proves that the total capacitance change ΔC has a strictly linear relationship with the liquid level height h .

Exercise

A variable dielectric displacement sensor consists of two square metal plates of side 5 cm, separated by a distance of 1 mm. A sheet of dielectric material of thickness 1 mm and the same area as the plates can slide between them. Given that the relative permittivity of air is 1.0 and that of the dielectric is 4.0, calculate the capacitance of the sensor for displacements x of 0.0, 2.5, and 5.0 cm.

- Plate length/width: $w = 5 \text{ cm} = 0.05 \text{ m}$
- Plate separation: $d = 1 \text{ mm} = 0.001 \text{ m}$
- Permittivity of free space: $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
- Relative permittivity of air: $\epsilon_{r1} = 1.0$
- Relative permittivity of dielectric: $\epsilon_{r2} = 4.0$



Solution

The basic formula for a flat parallel-plate capacitor is:

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

The total area of the plate is $w \times w$. When the displacement is x , the area of the dielectric part is

$A_{dielectric} = w \cdot x$, and the area of the air part is $A_{air} = w \cdot (w - x)$.

Therefore, the total capacitance C_{total} is:

$$C_{total} = C_1 + C_2 = \frac{\varepsilon_0 \varepsilon_{r2}(w \cdot x)}{d} + \frac{\varepsilon_0 \varepsilon_{r1}(w \cdot (w - x))}{d}$$

$$C_{total} = \frac{\varepsilon_0 w}{d} [\varepsilon_{r2} x + \varepsilon_{r1}(w - x)]$$

$$C_{total} = 4.425 \times 10^{-10} \cdot (3.0x + 0.05)$$

- **Case 1: When displacement $x = 0.0 \text{ cm} = 0.0 \text{ m}$** (The space is entirely filled with air)

$$C = 4.425 \times 10^{-10} \cdot (3.0 \times 0 + 0.05)$$

$$C = 4.425 \times 10^{-10} \times 0.05 = 2.2125 \times 10^{-11} \text{ F} = \mathbf{22.13 \text{ pF}}$$

- **Case 2: When displacement $x = 2.5 \text{ cm} = 0.025 \text{ m}$** (The space is exactly half air and half dielectric)

$$C = 4.425 \times 10^{-10} \cdot (3.0 \times 0.025 + 0.05)$$

$$C = 5.53125 \times 10^{-11} \text{ F} = \mathbf{55.31 \text{ pF}}$$

- **Case 3: When displacement $x = 5.0 \text{ cm} = 0.05 \text{ m}$** (The space is entirely filled with the dielectric sheet)

$$C = 4.425 \times 10^{-10} \cdot (3.0 \times 0.05 + 0.05)$$

$$C = 8.85 \times 10^{-11} \text{ F} = \mathbf{88.50 \text{ pF}}$$

Exercise

A parallel plate capacitive pressure sensor consists of two circular plates of diameter 2 cm separated by an air gap of 1 mm. Given that the relative permittivity of air is 1.0 and the permittivity of free space $\epsilon_0 = 8.85 \text{ pF/m}$, calculate the initial capacitance of the sensor.

- Diameter (D): $2 \text{ cm} = 0.02 \text{ m}$
- Radius (r): $\frac{D}{2} = 1 \text{ cm} = 0.01 \text{ m}$
- Initial separation distance (d): $1 \text{ mm} = 0.001 \text{ m} = 1 \times 10^{-3} \text{ m}$
- Permittivity of free space (ϵ_0): $8.85 \text{ pF/m} = 8.85 \times 10^{-12} \text{ F/m}$
- Relative permittivity of air (ϵ_r): 1.0

Solution

The core governing formula for the capacitance of a flat parallel-plate capacitor is:

$$C_0 = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

$$A = \pi r^2 = \pi \times (0.01 \text{ m})^2 = 0.0001\pi \text{ m}^2 = \pi \times 10^{-4} \text{ m}^2$$

$$A \approx 3.1416 \times 10^{-4} \text{ m}^2$$

$$C_0 = \frac{(8.85 \times 10^{-12} \text{ F/m}) \times 1.0 \times (3.1416 \times 10^{-4} \text{ m}^2)}{1 \times 10^{-3} \text{ m}}$$

Convert the final answer back to picofarads (pF), noting that $1 \text{ pF} = 10^{-12} \text{ F}$:

$$C_0 \approx \mathbf{2.78 \text{ pF}}$$

Core Essentials of Strain Gauges

Core Conversion Chain and Physical Formulas

- **Physical Process:** External Force $\rightarrow$ Structural Stress (σ) $\rightarrow$ Material Strain (ε) $\rightarrow$ Resistance Change ($\Delta R/R$)
- **Mechanics Formula (Hooke's Law):**

$$\varepsilon = \frac{\sigma}{E}$$

- **Electronics Formula (Strain Effect):**

$$\frac{\Delta R}{R} = K \cdot \varepsilon$$

(Note: K is the sensitivity coefficient of the strain gauge, usually around 2.0)

- Strain is extremely small. Problems often give microstrain ($\mu\varepsilon$), which must be converted during calculation: **$1\mu\varepsilon = 10^{-6}$** . The standard unit for Young's modulus E and stress σ is Pa or N/m^2 .

DC Wheatstone Bridge

- **Purpose:** To convert the extremely small resistance change (ΔR) into a measurable voltage output (U_{out}).
- **Comparison of Three Classic Bridge Configurations:**
 - **Quarter Bridge - 1 gauge:**

$$U_{out} \approx \frac{1}{4} U_{ex} K \varepsilon$$

- **Half Bridge (Differential) - 2 gauges, one in tension and one in compression:**

$$U_{out} \approx \frac{1}{2} U_{ex} K \varepsilon$$

- **Full Bridge (Differential) - 4 gauges, two in tension and two in compression:**

$$U_{out} = U_{ex} K \varepsilon$$

Exercise

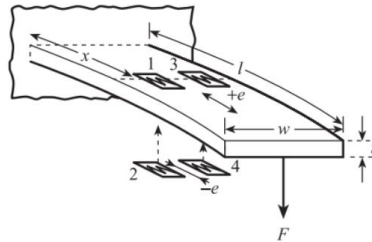
Four strain gauges are bonded onto a cantilever as shown in Figure 8.20(a). Given that the gauges are placed halfway along the cantilever and the cantilever is subject to a downward force of 0.5 N, use the data given below to calculate the resistance of each strain gauge:

Cantilever data

Length $l = 25 \text{ cm}$
Width $w = 6 \text{ cm}$
Thickness $t = 3 \text{ mm}$
Young's modulus $E = 70 \times 10^9 \text{ Pa}$

Strain gauge data

Gauge factor $G = 2.1$
Unstrained resistance $R_0 = 120 \Omega$



(a)

Solution

Step 1: Calculate the bending moment (M) at the location of the strain gauges

The force F is applied at the free end of the cantilever, and the strain gauges are placed *halfway* along its length. Therefore, the distance (moment arm) from the applied force to the gauges is $x = \frac{l}{2}$.

- Force $F = 0.5 \text{ N}$

- Distance $x = \frac{25 \text{ cm}}{2} = 12.5 \text{ cm} = 0.125 \text{ m}$

$$M = F \times x = 0.5 \times 0.125 = \mathbf{0.0625 \text{ Nm}}$$

Step 2: Calculate the second moment of area (I) of the cantilever cross-section

For a rectangular cross-section, the formula is $I = \frac{w \cdot t^3}{12}$, where w is the width and t is the thickness.

- Width $w = 6 \text{ cm} = 0.06 \text{ m}$

- Thickness $t = 3 \text{ mm} = 0.003 \text{ m}$

$$I = \frac{0.06 \times (0.003)^3}{12} = \frac{0.06 \times 2.7 \times 10^{-8}}{12} = \mathbf{1.35 \times 10^{-10} \text{ m}^4}$$

Step 3: Calculate the stress (σ) at the surface of the cantilever

The stress at the surface (distance $y = \frac{t}{2}$ from the neutral axis) is given by the bending stress formula $\sigma = \frac{M \cdot y}{I}$.

- Distance to surface $y = \frac{0.003 \text{ m}}{2} = 0.0015 \text{ m}$

$$\sigma = \frac{0.0625 \times 0.0015}{1.35 \times 10^{-10}} = \mathbf{694,444.44 \text{ Pa}}$$

Solution

Step 4: Calculate the strain (ε) experienced by the gauges

Using Hooke's Law ($\sigma = E \cdot \varepsilon$), we can find the strain.

- Young's modulus $E = 70 \times 10^9$ Pa

$$\varepsilon = \frac{\sigma}{E} = \frac{694444.44}{70 \times 10^9} \approx \mathbf{9.92 \times 10^{-6}}$$

Step 5: Calculate the change in resistance (ΔR) for each gauge

The relationship between strain and resistance change is given by the gauge factor $G = \frac{\Delta R/R_0}{\varepsilon}$.

- Gauge factor $G = 2.1$

- Unstrained resistance $R_0 = 120 \Omega$

$$\Delta R = R_0 \cdot G \cdot \varepsilon$$

$$\Delta R = 120 \times 2.1 \times (9.9206 \times 10^{-6}) = \mathbf{0.0025 \Omega}$$

Step 6: Determine the final resistance of each strain gauge

When a downward force is applied to a cantilever, the top surface stretches (tension) and the bottom surface compresses (compression).

- For the two gauges on the **top surface (Tension)**, the resistance increases:

$$R_{top} = R_0 + \Delta R = 120 + 0.0025 = \mathbf{120.0025 \Omega}$$

- For the two gauges on the **bottom surface (Compression)**, the resistance decreases:

$$R_{bottom} = R_0 - \Delta R = 120 - 0.0025 = \mathbf{119.9975 \Omega}$$

Exercise

A strain gauge having an unstrained resistance of $120\ \Omega$ and a gauge factor of 2.1 is bonded onto a steel girder so that it experiences a tensile stress of 10^8 Pa . If Young's modulus for steel is $2 \times 10^{11}\text{ Pa}$, calculate the strained resistance of the gauge.

Solution

- Unstrained resistance (R_0) = $120\ \Omega$
- Gauge factor (G) = 2.1
- Tensile stress (σ) = $10^8\ \text{Pa}$
- Young's modulus for steel (E) = $2 \times 10^{11}\ \text{Pa}$

Calculate the Strain (ε)

$$\varepsilon = \frac{\sigma}{E}$$

$$\varepsilon = \frac{10^8\ \text{Pa}}{2 \times 10^{11}\ \text{Pa}} = 0.5 \times 10^{-3} = \mathbf{5 \times 10^{-4}}$$

Calculate the Change in Resistance (ΔR)

$$\Delta R = R_0 \cdot G \cdot \varepsilon$$

$$\Delta R = 120 \times 2.1 \times (5 \times 10^{-4})$$

$$\Delta R = 252 \times 0.0005 = \mathbf{0.126\ \Omega}$$

Calculate the Final Strained Resistance

$$R_{\text{strained}} = R_0 + \Delta R$$

$$R_{\text{strained}} = 120 + 0.126 = \mathbf{120.126\ \Omega}$$